

The Day Began at Eleven at Night: The Double-Hour and Its Measure

Chujia Kuang (邝楚嘉)

嘉言一得 (Jiayan Yide)

Independent Researcher, Jieyang, Guangdong, China

2026-10-04 · 预印本 (preprint) · 未经同行评议 · v3

DOI: 10.5281/zenodo.22866122

Chujia Kuang (邝楚嘉)

Independent Researcher, Jieyang, Guangdong, China

Corresponding author: Chujia Kuang, Independent Researcher, Jieyang, Guangdong, China. E-mail: chirca@163.com

Abstract

The day, for most people, begins at daybreak or at zero o'clock. For the ancients it began at eleven at night. This note sets out why: the double-hour 子 (zi), the first of the twelve, is the moment of the day farthest from the sun, the hour of greatest yin — and it is exactly there that the day is held to turn, just as the winter solstice, the day of the year farthest from the sun, is the point at which the single yang returns. The twelve double-hours are then presented as a set of names drawn not from an abstract scale but from the ordinary round of the day; two of the twelve are mealtimes, which is itself evidence that the ancients ate twice a day. A double-hour is exactly two hours, and after the Tang each was divided into an initial (初, chu) and a medial (正, zheng) half, giving twenty-four divisions whose length matches the modern hour — the origin of the word xiaoshi, "small hour". The note traces the twelvefold division against the rival schemes of two, five and ten divisions, and dates its final form to the Taichu calendar reform of 104 BCE; it then gives the length of the ke (刻) under the hundred-ke and ninety-six-ke systems (14.4 minutes before 1670, 15 minutes after) and the five watches (更, geng) of the night. It closes by noting that the double-hour was measured by the sundial, and that clock time and solar time are not the same thing. The note is descriptive; it contains no predictive claims.

Keywords: double-hour; 子 hour; midnight; twelve double-hours; ke; five watches; sundial; solar time

1. Why the starting point is at midnight

What time does a day begin?

Most people would answer: at daybreak. Or at zero o'clock.

The ancients' answer is stranger — eleven at night.

This was not chosen at random. Behind it lies a whole understanding of time; once this point is grasped, the twelve double-hours that follow will stand.

First, a counter-intuitive fact: of the twelve double-hours, the first, 子 (zi), is the moment of the day farthest from the sun.

The 子 hour runs from 23:00 to 01:00; the sun has moved to the far side of the earth, directly beneath our feet. It is the time of day when the yin is at its height.

Yet the ancients set the beginning of the day precisely at this darkest moment.

Why?

The *Sanming tonghui*, in its chapter "On the Double-Hours", gives an explanation of remarkable aptness:

若子时，则上半时在夜半前，属昨日。下半时在夜半后，属今日。亦犹冬至得十一月中气，一阳来复，为天道之初耳。（In the case of the 子 hour, the first half falls before midnight and belongs to the previous day; the second half falls after midnight and belongs to the present day. It is just as the winter solstice, which takes the second qi of the eleventh month, when the single yang returns, is the beginning of the way of heaven.）

The meaning is: the 子 hour stands to a day as the winter solstice stands to a year.

The winter solstice is the day of the year with the shortest daylight and the greatest yin, but it is also the turning point at which the yang begins to rise again — the saying "at the winter solstice the single yang is born". The 子 hour is the same: it is the darkest moment of the day, and it is the moment at which the yang begins to return to life.

So the *Leijing tuyi* says: "子者阳生之初" — "the 子 is the beginning of the birth of the yang."

The beginning of the day lies not at the brightest point but at the instant of turning, at the darkest point. This way of thinking is of one heart with the saying "when adversity reaches its extreme, good fortune comes".

Two corroborations:

- Li Chunfeng of the Tang records in the *Xin Tangshu*, "Treatise on the Calendar": "古历分日，起于子半" — "the old calendars divided the day, beginning at the middle of the 子 [hour]" — taking the midpoint of the 子 hour, that is, zero o'clock, as the boundary of the day.
- Fu Sheng of the early Han, in the *Shangshu dazhuan*: "周以十一月为正，色尚赤，以夜半为朔" — "the Zhou took the eleventh month as the corrected beginning, favoured the colour red, and

took midnight as the first of the month" — the Zhou took midnight as the beginning of a month.

Thus there are three "beginnings", each answering to the next:

The beginning of the year is zhengyue; the beginning of a month is the first day; the beginning of a day is the middle of the 子 hour.

A widely circulated statement should be corrected in passing: "the 子 hour is twelve o'clock at midnight" is not accurate.

Twelve o'clock is called 子正 (zizheng), the midpoint of the 子 hour; the 子 hour itself is 23:00 to 01:00, a full two hours.

By the same token, the 午 hour is 11:00 to 13:00, and it is 午正 (wuzheng) that is twelve o'clock.

On a chart this distinction makes a real difference: a birth at half past eleven at night and a birth at half past midnight are separated by only one hour, yet fall respectively in 子初 and 子正.

2. The twelve double-hours are twelve pictures of daily life

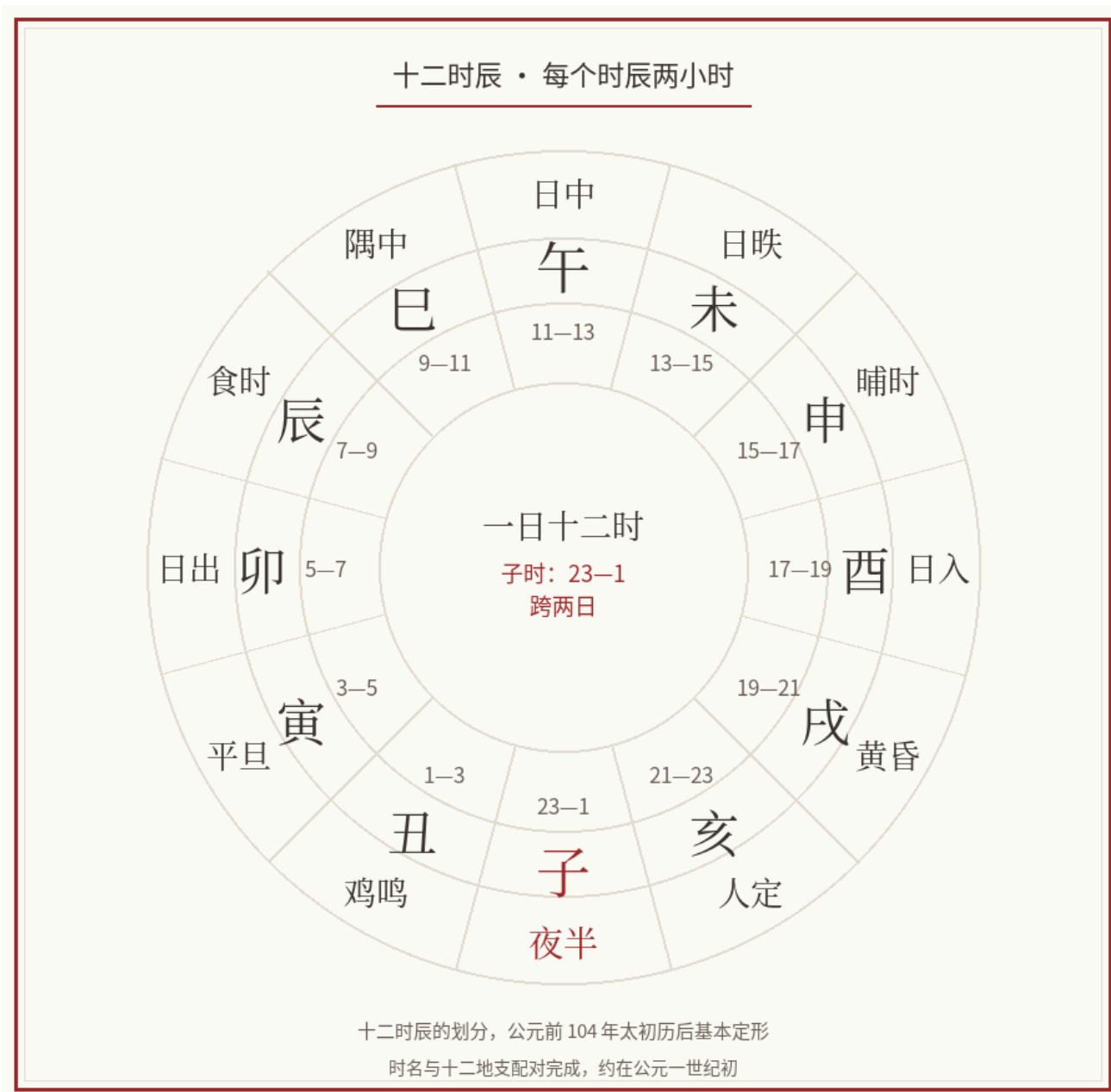
The ancients did not first have "子, 丑, 寅, 卯" and then fit them to time. They first had the daily round, and then gave each stretch a name.

Double-hour	Clock	The picture within the name
子	23—1	Midnight (夜半, yeban). 姑苏城外寒山寺, 夜半钟声到客船 — "Outside Gusu city, at Cold Mountain Temple, the midnight bell reaches the traveller's boat"
丑	1—3	Cock-crow (鸡鸣, jiming). 鸡声茅店月, 人迹板桥霜 — "The cock's cry, the moon over a thatched inn; footprints, frost on a plank bridge"
寅	3—5	Dawn (平旦, pingdan). 夜与日交替之际, 《孟子》说的「平旦之气」 — "the hour when night and day alternate; the 'dawn air' spoken of in the Mencius"
卯	5—7	Sunrise (日出, richu). 日出而作, 日入而息 — "out at sunrise and in at sunset"
辰	7—9	Meal time (食时, shishi). 吃早饭的时候 — the time of the morning meal
巳	9—11	Forenoon (隅中, yuzhong). 临近中午 — drawing near to noon
午	11—13	Midday (日中, rizhong). 太阳升到中天 — the sun rises to mid-heaven
未	13—15	The sun's decline (日昃, ridie). 日头开始偏西 — the sun begins to lean west
申	15—17	Late-afternoon meal (晡时, bushi). 吃第二顿饭的时候 — the time of the second meal
酉	17—19	Sunset (日入, riru). 山中相送罢, 日暮掩柴扉 — "the farewells in the hills are done; at dusk the brushwood gate is closed"
戌	19—21	Dusk (黄昏, huanghun). 月上柳梢头, 人约黄昏后 — "the moon rises above the willow tops; a tryst is kept after dusk"
亥	21—23	People at rest (人定, rending). 奄奄黄昏后, 寂寂人定初 — "after the fading dusk, at the first still hour when people settle"

See it now — of the twelve names, two are mealtimes: 辰 is "meal time", the morning meal, and 申 is the "late-afternoon meal", the second meal.

This shows that the ancients ate only twice a day. One meal between seven and nine in the morning, one between three and five in the afternoon, with nothing in between.

This detail is more useful than memorising the twelve earthly branches. It reminds you that the double-hours did not drop from the sky as an abstract scale; they grew out of daily life.



3. A double-hour is exactly two hours

The twelve double-hours divide a day and night evenly into twelve parts, so one double-hour equals two hours — a day of exactly twenty-four hours.

This "twenty-four" is no accident. From the Tang onward each double-hour was divided into an initial (初) and a medial (正) half of one hour each, so that the twelve double-hours became twenty-four parts. This is China's own "twenty-four hours to the day", and each part so produced is exactly equal in length to today's hour.

More interesting still is the origin of the word "small hour" (xiaoshi, 小时).

Qian Daxin of the Qing, in the *Shijiazhai yangxin lu*, "The Twenty-Four Hours", puts it clearly:

一日分十二时，每时又分为二，曰「初」，曰「正」，是为二十四小时。(A day is divided into twelve double-hours, and each double-hour is again divided into two, called the initial and the medial, giving the twenty-four hours.)

The ancients called a double-hour a **great hour** (大时, dashi). Later, to keep step with the internationally current twenty-four-hour system, each great hour was split in two, and the half so produced was called the **small hour** (小时, xiaoshi).

Half of a great hour is a small hour.

We say the word every day, and few have wondered why it carries the character "small" — small by contrast with the "great hour" that is the double-hour.

4. Why twelve parts and not ten

How many parts to cut the day into — the ancients tried more than one scheme.

The crudest division is into two: sunrise to sunset is "day", sunset to sunrise is "night", called the "two-part system of dark and light". This does not even amount to a unit, since the length of day and night changes with the seasons.

The Chunqiu Zuo zhuan, under the fifth year of Duke Zhao, has a further line: "日之数十，故有十时" — "the numbers of the sun are ten, and so there were ten hours." This shows that there was once a ten-double-hour system, dividing the day into ten parts.

There was later also a five-double-hour system.

What won in the end was twelve. The reasons are not hard to see:

- A year has twelve lunar months, and twelve was already the period the ancients knew best.
- The year-star (Jupiter) takes about twelve years to circuit the sky, and the ancients used it to record the years.
- The twelve earthly branches form a set of their own, and can be made to mark direction, month and double-hour alike.

The correspondence of double-hour to direction is equally direct: 子 is true north, 午 is true south.

The sun rises to its highest point in the true south at the 午 hour (upper culmination), and sinks to its lowest in the true north at the 子 hour (lower culmination). So "midday" is 午 and "midnight" is 子 — the names of the double-hours correspond directly to the sun's position in the sky.

5. When it was settled

- The Zhouli already has the phrase "十有二辰" — "the twelve 辰"; the shape is visible in the Shang and Zhou.
- In 104 BCE, after Emperor Wu of Han promulgated the Taichu calendar, it took substantially its final form.
- Zhao Yi of the Qing, in the Gaiyu congkao: "In the Taichu period of the Han the Taichu calendar was put into effect; '其以一日分十二时，而以干支为纪' — 'it divided a day into twelve double-hours and recorded them by the stems and branches'."
- The pairing of the twelve names (midnight, cock-crow, ...) with the twelve earthly branch names was completed only about the beginning of the first century CE.

So the statement "the twelve double-hours have existed since antiquity" is not accurate. The names and the branches were matched only two thousand years ago.

6. How long, exactly, is a ke

The ancients measured time with a water-clock (a bronze clepsydra), dividing a day and night into one hundred ke (刻); the system is thought to have existed already in the Shang.

A day and night is $1,440 \text{ minutes} \div 100 \text{ ke} = 14.4 \text{ minutes to the ke}$, that is, 14 minutes 24 seconds.

The trouble is that one hundred ke cannot be divided evenly by twelve: each double-hour comes to $8\frac{1}{3}$ ke, which is very inconvenient to reckon.

In the ninth year of Kangxi of the Qing (1670) the hundred ke were changed to ninety-six ke, giving "twelve double-hours to the day, eight ke to the double-hour, fifteen minutes to the ke, sixty seconds to the minute" — each ke exactly 15 minutes, in step with the internationally current hours, minutes and seconds.

So is a "ke" 14.4 minutes or 15 minutes? Both are right; it depends on the dynasty: 14.4 before Kangxi, 15 after.

In passing, a widely circulated statement should be laid aside —

"午时三刻" is not twelve o'clock noon.

By the hundred-ke system: the initial of the 午 hour begins at 11:00, and three ke = $3 \times 14.4 \text{ minutes} \approx 43.2 \text{ minutes}$.

"午时三刻" is about 11:43.

When a novel says "execution at 午时三刻", it means around twenty to twelve in the morning, not noon. The notion that this is "when the yang is at its height" likewise derives from this double-hour.

7. Why "the third watch" and "酉正三刻" differ so much

At night there was another clock: the night was divided into five watches (更, geng), each announced by a drum, and each watch again into five points (点, dian).

Watch	Equivalent today
First watch	19:12
Second watch	21:36
Third watch	0:00
Fourth watch	2:24
Fifth watch	4:48

One night is forty ke (daytime being sixty ke), each watch about 144 minutes, each point about 28.8 minutes.

Here there is a dividing line of period: before the Song the watch-and-point system was used; after the Song the double-hour-and-ke system.

So when the Shuihu zhuan says "at the third watch" and the Honglou meng says "酉正三刻", this is not the author writing at random; it is the difference of period between two systems of reckoning time.

8. One last thing easily overlooked

These double-hours were measured by the ancients with a sundial (日晷).

The sundial reads the sun's actual position: the moment the sun reaches its highest point in the true south is the local "medial of the 午 hour".

The twelve o'clock on our clocks today is another matter altogether.

The same two characters "午时" point, for the ancients, to one moment, and for today's clock to another — and across most of China the two do not agree; in some places they are two hours apart.

This is for later.

Closing

The beginning of the day is at midnight, not at daybreak.

Once this is grasped, the twelve double-hours are no longer twelve cold graduations — where the sun goes, that is what people should be doing; this scale grew out of the soil.

What comes next is zhengyue — which was not originally the first month.

Declarations

Funding. This research received no external funding.

Competing interests. The author declares no competing interests.

Data availability. No new data were created or analysed in this study.

Language. The original Chinese text is deposited as a companion file to this record (古人的一天，从晚上十一点开始).

References

- [1] Meng Ke. Mengzi [孟子][M]. Pre-Qin period.
- [2] Zuozhuan [左传][M]. Pre-Qin period.
- [3] Zhouli [周礼][M]. Pre-Qin period.
- [4] Fu Sheng. Shangshu dazhuan [尚书大传][M]. Western Han.
- [5] Ouyang Xiu, Song Qi. Xin Tangshu [新唐书][M]. Song dynasty.
- [6] Wan Mingying. Sanming tonghui [三命通会][M]. Ming dynasty.
- [7] Zhang Jiebin. Leijing tuyi [类经图翼][M]. Ming dynasty.
- [8] Shi Nai'an. Shuihu zhuan [水浒传][M]. Ming dynasty.
- [9] Qian Daxin. Shijiazhai yangxin lu [十驾斋养新录][M]. Qing dynasty.
- [10] Zhao Yi. Gaiyu congkao [陔余丛考][M]. Qing dynasty.
- [11] Cao Xueqin. Honglou meng [红楼梦][M]. Qing dynasty.